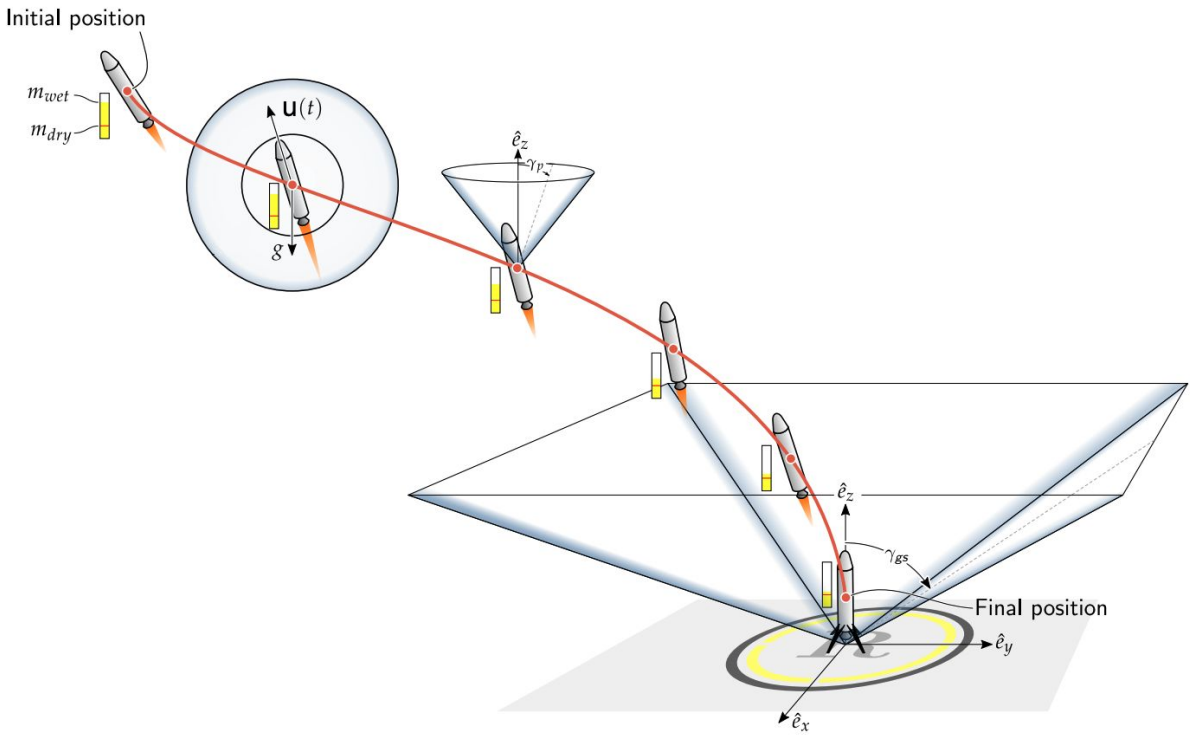
Advanced optimal control: from reusable rocket landing to efficient training of neural networks

IPSA course 2026

Riccardo Bonalli - riccardo.bonalli@cnrs.fr

Laboratoire des Signaux et Systèmes
CNRS and Université Paris-Saclay

Example: reusable rocket



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$$r = \text{position}$$
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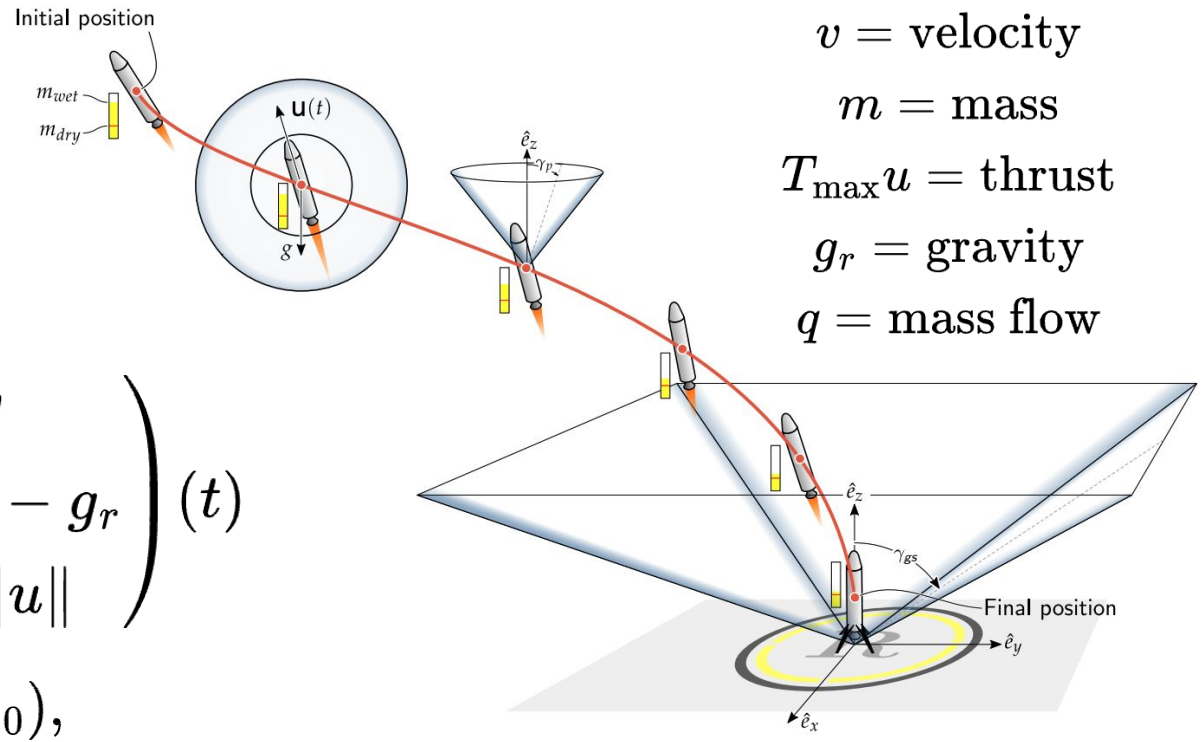
$$m = \text{mass}$$
$$T_{\max} u = \text{thrust}$$
$$g_r = \text{gravity}$$
$$q = \text{mass flow}$$



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$$(r, v, m)(0) = (r_0, v_0, m_0),$$
$$(r, v)(t_f) = (0, 0)$$

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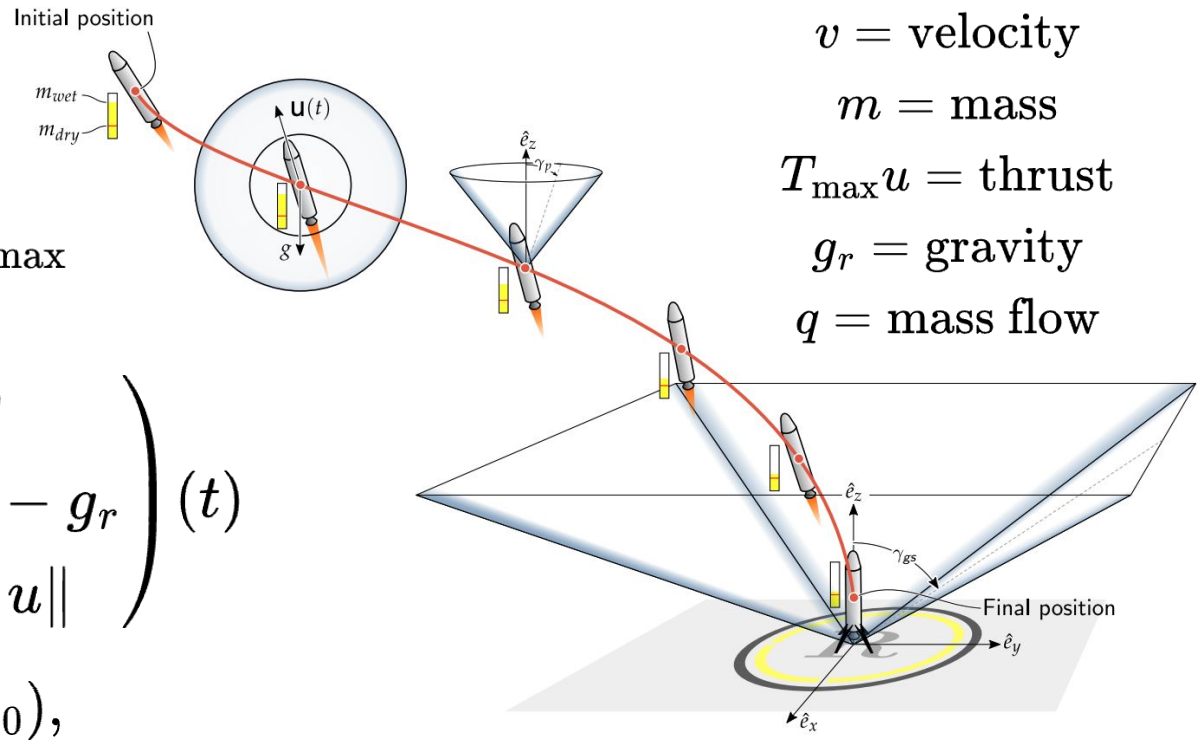

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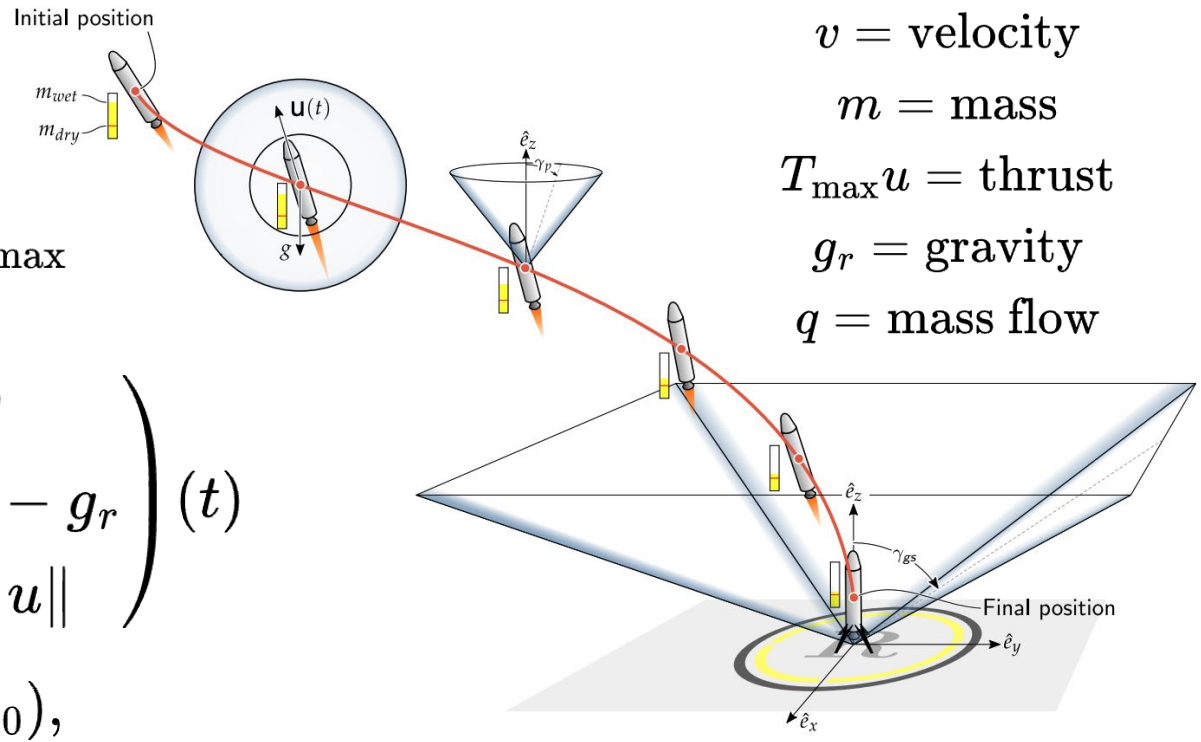

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Dynamics + initial/final conditions

$$\begin{cases} x(t) \in \mathbb{R}^n, \quad t \in [0, t_f], \\ \dot{x}(t) = f(t, x(t), u(t)), \\ x(0) = x_0, \quad g(x(t_f)) = 0 \end{cases}$$

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Example: reusable rocket

$$\min_{u} \; -m(t_f) \longrightarrow \min_{u(\cdot)\in L^\infty} \; h(x(t_f)) + \int_0^{t_f} f^0(t, x(t), u(t))\, dt$$

Cost (functional)

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Optimal Control Problem (OCP)

Will learn theory and algorithms
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1. OCP is infinite dimensional
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1. OCP is infinite dimensional (controls are in L infinity)
2. OCP is non-convex (dynamics and cost are non-convex)

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Ultimate goal

Get a comprehensive understanding of the key challenges faced by contemporary aerospace control engineers, addressing both theoretical and practical aspects

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5. Final presentation of the results (the SOLE evaluation step).

→ Group evaluation: please form groups of 2 / 3 people and <u>report them in the distributed paper</u>!

Course material

Lecture slides, codes, and exam information are fully available at:

https://rbonalli.github.io/ipsa

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Lecture 1 - Existence of solutions to OCP

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Optimal Control Problem (OCP)

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$$\min_{u(\cdot)\in L^\infty} \; h(x(t_f)) + \int_0^{t_f} f^0(t, x(t), u(t))\, dt \longrightarrow \begin{array}{c} h : \mathbb{R}^n \to \mathbb{R} \\ f^0 : \mathbb{R}_+ \times \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R} \end{array}$$

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Key facts

1. The final time t_f may be free
2. All mappings are continuous…

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Key facts

1. The final time t_f may be free
2. All mappings are continuous… except for controls!
3. $u : [0, t_f] \to U$ such that $u(\cdot) \in L^\infty([0, t_f], U)$

Today's detailed schedule

1. Revisit the concept of $L^{\infty}([0, t_f], U)$.

2. Revisit Ordinary Differential Equations (ODE): existence and uniqueness of solutions given non smooth controls.

3. Conditions for the existence of solutions to OCP and application to the reusable rocket landing problem.

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Quick reminders on Lp spaces - Theory

In this course we will often deal with:

$$L^\infty([0,t_f],U) \triangleq \{u : [0,t_f] \to U \subseteq \mathbb{R}^m \text{ s.t. } u(\cdot) \text{ is } \textit{measurable} \quad \text{and} \quad \|u(\cdot)\|_{L^\infty} < \infty\}$$

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1. "Basically", integrals like $\int_0^t \varphi(s,u(s))\,ds$ are well-defined for every $t\in[0,t_f]$ and $\varphi\in C^0([0,t_f]\times\mathbb{R}^m,\mathbb{R}^\ell)$

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2. The "supremum" norm $\|u(\cdot)\| \triangleq \sup_{t\in[0,t_f]} \|u(t)\|$ is well-defined and finite.

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$$m=1, \quad U\triangleq[-1,1]\subseteq\mathbb{R}, \quad t_{sw}\in[0,t_f]$$

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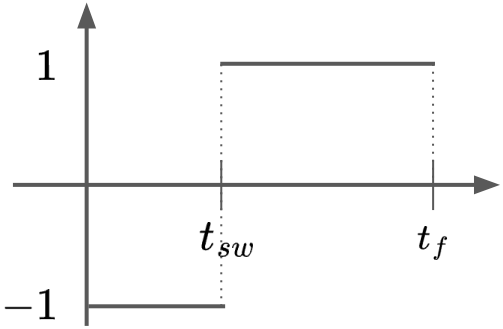
$$u_{t_{sw}}(t)\triangleq\begin{cases}-1, & t\le t_{sw}\\ 1, & t>t_{sw}\end{cases}, \quad t\in[0,t_f]$$



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2. We need to answer the following question:

Under what assumptions on f does a solution to ODE <u>exist uniquely</u>?

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3. <u>Assumptions ODE</u> - There exist $\bar{x} \in \mathbb{R}^n$ and a C^0 function $L(\cdot) > 0$ such that:

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$A.$ $\|f(t, \bar{x}, u)\| \leq L(\|u\|), \quad (t,u) \in [0,t_f] \times U \quad (u - boundness)$

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$A.$ $\|f(t, \bar{x}, u)\| \leq L(\|u\|), \quad (t,u) \in [0,t_f] \times U \quad (u - boundness)$

$B.$ $\|f(t,x,u) - f(t,y,u)\| \leq L(\|u\|)\|x-y\|, \quad (t,u) \in [0,t_f] \times U, \; x,y \in \mathbb{R}^n \quad (Lipschitzianity)$

Reminders on ODE - Theory

From now on: $u(\cdot) \in L^\infty([0,t_f],U), \quad U \subseteq \mathbb{R}^m$ (it might hold $U = \mathbb{R}^m$)

In this course we will often deal with:

ODE $\begin{cases} \dot{x}(t) = f(t, x(t), u(t)), \\ t \in [0, t_f], \quad x(0) = x_0 \end{cases}$ where $f : \mathbb{R}_+ \times \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$ is "at least" continuous

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4. <u>Theorem ODE</u> (Existence and uniqueness of solutions to ODE).

Under <u>Assumptions ODE</u>, ODE has a unique solution $x : [0,t_f] \to \mathbb{R}^n$.

Reminders on ODE - Examples

Double integrator - Used in electronic circuit modeling

$$\begin{cases} \dot{y}(t) = z(t) \\ \dot{z}(t) = u_{t_{sw}}(t) \\ t \in [0, t_f],\ x(0) = y(0) = 0 \end{cases}$$

where $u_{t_{sw}}(t) \triangleq \begin{cases} -1, & t \leq t_{sw} \\ 1, & t > t_{sw} \end{cases}, \quad t \in [0, t_f]$

(clearly $u_{t_{sw}} \in L^{\infty}([0, t_f], [-1, 1])$)

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$$n = 2, m = 1, \quad x \triangleq (y, z), \quad f(t, x, u) \triangleq \begin{pmatrix} z \\ u \end{pmatrix}$$

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$$\|f(t, 0, u)\| = |u| \quad \text{and} \quad \|f(t, x_1, u) - f(t, x_2, u)\| = \left\| \begin{pmatrix} z_1 - z_2 \\ 0 \end{pmatrix} \right\|$$
$$= |z_1 - z_2| \leq \|x_1 - x_2\|$$

Reminders on ODE - Examples

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Assumptions ODE hold: Theorem ODE applies!

Reminders on ODE - Examples

Reusable rocket dynamics

$$\frac{d}{dt}\begin{pmatrix} r \\ v \\ m \end{pmatrix}(t) = \begin{pmatrix} v \\ \frac{T_{\max} u}{m} - g_r \\ -q\|u\| \end{pmatrix}(t)$$ with controls $u(\cdot) \in L^\infty([0, t_f], U)$,

$$U \triangleq \{u \in \mathbb{R}^3 : \ 0 < u_{\min} \leq \|u\| \leq u_{\max}\}$$

$$(r, v, m)(0) = (r_0, v_0, m_0)$$

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<u>Solution</u>: See the blackboard for details.

$$\|f(t, (0, 0, m_{\min}), u)\| \leq \|g_r\| + \left(q + \frac{T_{\max}}{m_{\min}}\right)\|u\| \quad \text{and}$$

$$\|f(t, x_1, u) - f(t, x_2, u)\| \leq \max\left(1, \frac{T_{\max}\|u\|}{m_{\min}^2}\right)\|x_1 - x_2\| \longrightarrow$$ <u>Theorem ODE</u> applies!

Today's detailed schedule

1. Revisit the concept of $L^\infty([0, t_f], U)$.

2. Revisit Ordinary Differential Equations (ODE): existence and uniqueness of solutions given non smooth controls.

3. Conditions for the existence of solutions to OCP and application to the reusable rocket landing problem.

Existence of solutions to OCP - Theory

Cost (functional)

$$\min_{u(\cdot)\in L^\infty} \; h(x(t_f)) + \int_0^{t_f} f^0(t, x(t), u(t))\, dt$$

(Control) constraints

$$u(t) \in U \subseteq \mathbb{R}^m, \quad t \in [0, t_f]$$

Dynamics + initial/final conditions

$$\begin{cases} x(t) \in \mathbb{R}^n, \quad t \in [0, t_f], \\ \dot{x}(t) = f(t, x(t), u(t)), \\ x(0) = x_0, \quad g(x(t_f)) = 0 \end{cases}$$

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Maybe more than one!!!

Existence of solutions to OCP - Theory

<u>Assumptions Existence OCP</u>

Existence of solutions to OCP - Theory

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1. The control set $U \subseteq \mathbb{R}^m$ is compact and the final time t_f is fixed

Existence of solutions to OCP - Theory

Can be relaxed…

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1. The control set $U \subseteq \mathbb{R}^m$ is compact and the final time t_f is fixed
2. All the mappings $h, f^0, f, \text{ and } g$ are C^1, and <u>Assumptions ODE</u> hold

Existence of solutions to OCP - Theory

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3. The epigraphs of extended velocities are <u>convex</u> for every $(t, x) \in [0, t_f] \times \mathbb{R}^n$:

$$V(t,x) \triangleq \left\{ \begin{pmatrix} f(t,x,u) \\ f^0(t,x,u) + \gamma \end{pmatrix} : \; u \in U, \; \gamma \geq 0 \right\} \subseteq \mathbb{R}^{n+1}$$

Existence of solutions to OCP - Theory

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<u>Theorem Existence OCP</u> (Existence of solutions to OCP).

Under the previous <u>Assumptions Existence OCP</u>, a feasible OCP has <u>at least one</u> optimal solution $u^*(\cdot) \in L^\infty([0, t_f], U)$, with optimal trajectory $x_{u^*} : [0, t_f] \to \mathbb{R}^n$.

Existence of solutions to OCP - Examples

Minimal energy double integrator - Used in electronic circuit eco-phasing

$$\min_{u(\cdot)\in L^\infty([0,t_f],[-1,1])} \int_0^{t_f} u(t)^2\, dt, \quad t_f > 0 \text{ fixed}$$

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"Reach the origin with zero final velocity and minimal energy"

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By assuming <u>OCP is feasible</u>, let us check <u>Assumptions Existence OCP</u> hold:

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By assuming <u>OCP is feasible</u>, let us check <u>Assumptions Existence OCP</u> hold:

1. The control set $U = [-1, 1]$ is compact (and the final time is clearly fixed)

2. We already checked <u>Assumptions ODE</u> hold. In addition, by denoting $x \triangleq (y, z)$ the mappings $h(x) = 0, \;\; f^0(t, x, u) = u^2, \;\; f(t, x, u) = (z, u), \;\; \text{and} \;\; g(x) = x$ are C^1

Existence of solutions to OCP - Examples

3. It remains to show the convexity of the following sets, for every $(t, x) \in [0, t_f] \times \mathbb{R}^2$:

$$V(t,x) = \left\{ \begin{pmatrix} f(t,x,u) \\ f^0(t,x,u) + \gamma \end{pmatrix} = \begin{pmatrix} z \\ u \\ u^2 + \gamma \end{pmatrix} : \; u \in [-1,1], \; \gamma \geq 0 \right\} \subseteq \mathbb{R}^3$$

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<u>Proof</u>: Fix $(t,x) \in [0,t_f] \times \mathbb{R}^2$, and let $\lambda \in (0,1)$, $\begin{pmatrix} z \\ u_1 \\ u_1^2 + \gamma_1 \end{pmatrix}, \begin{pmatrix} z \\ u_2 \\ u_2^2 + \gamma_2 \end{pmatrix} \in V(t,x)$

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Then $u_3 \triangleq \lambda u_1 + (1-\lambda) u_2 \in [-1,1]$

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Then $u_3 \triangleq \lambda u_1 + (1-\lambda) u_2 \in [-1,1]$, and since the function $u \mapsto u^2$ is convex, we have that $\gamma_3 \triangleq \lambda \gamma_1 + (1-\lambda)\gamma_2 + (\lambda u_1^2 + (1-\lambda) u_2^2 - u_3^2) \geq 0$

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Then $u_3 \triangleq \lambda u_1 + (1-\lambda) u_2 \in [-1,1]$, and since the function $u \mapsto u^2$ is convex, we have that $\gamma_3 \triangleq \lambda \gamma_1 + (1-\lambda)\gamma_2 + (\lambda u_1^2 + (1-\lambda) u_2^2 - u_3^2) \geq 0$

Thus

$$\lambda \begin{pmatrix} z \\ u_1 \\ u_1^2 + \gamma_1 \end{pmatrix} + (1-\lambda) \begin{pmatrix} z \\ u_2 \\ u_2^2 + \gamma_2 \end{pmatrix} = \begin{pmatrix} z \\ u_3 \\ u_3^2 + \gamma_3 \end{pmatrix} \in V(t,x)$$

Existence of solutions to OCP - Examples

3. It remains to show the convexity of the following sets, for every $(t,x) \in [0,t_f] \times \mathbb{R}^2$:

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<u>Theorem Existence OCP</u> applies: if feasible, this OCP has at least one solution!

Existence of solutions to OCP - Examples

Original problem:

Minimal fuel use reusable rocket landing

$$\min_u \; -m(t_f)$$

$$0 < u_{\min} \leq \|u(t)\| \leq u_{\max}$$

$$\frac{d}{dt}\begin{pmatrix} r \\ v \\ m \end{pmatrix}(t) = \begin{pmatrix} v \\ \frac{T_{\max} u}{m} - g_r \\ -q\|u\| \end{pmatrix}(t)$$

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$$\min_{u \in L^\infty([0,t_f],U)} \int_0^{t_f} \|u(t)\|^2 \, dt, \quad t_f > 0 \text{ fixed}$$

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Some fixed mass!

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<u>Solution</u>: See the blackboard for details.

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Existence of solutions to OCP - Examples

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1. The control set is clearly compact and the final time is clearly fixed
2. We already checked <u>Assumptions ODE</u> hold. Also, by denoting $x \triangleq (r,v) \in \mathbb{R}^6$

the mappings $h(x) = 0,\ f^0(t,x,u) = \|u\|^2,\ f(t,x,u) = \begin{pmatrix} v(t) \\ \frac{T_{\max} u(t)}{\bar{m}} - g_r \end{pmatrix},\ g(x) = x$ are C^1

Existence of solutions to OCP - Examples

Minimal energy reusable rocket landing

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3. Replicate the argument in the previous example.

Course schedule

1. Optimal Control Problems (OCP): review of ordinary differential equations; existence of solutions to OCP.

2. Optimality Conditions for OCP: the Maximum Principle and structure of optimal controls; application to reusable rocket landing (next lecture).

3. Python Session 1: real-world reusable rocket landing.

4. Python Session 2: learning dynamics through NeuralODE.

5. Final presentation of the results.

End of lecture 1

Questions?
Otherwise, see you in two weeks!